SULZER	Machine Operating Procedure (MOP) CHIEFTAIN 1860 Lathe Machine	Risk Assessment RA XXXXX

Issued By:	Prakash Giri	Region:	INMEC
Authorised By:		Sub-Region	Middle East
Applicable To:	Machine Operations	Service Centre:	SASOS Abu Dhabi

WARNING This document is a generic workshop SOP. Verify it against the exact CU 1250 machine manual, installed guarding, chuck/workholding system, accessories and your site risk assessment before controlled issue. Only trained and authorised persons may operate the lathe.

PPE requirements

	• Safety glasses • Single piece coveralls (Short sleeves) • Safety footwear
	• Gloves must not be worn near rotating equipment • Cut-resistant gloves must be worn when handling swarf, tooling, and when loading or unloading components onto or from the lathe • Chemical-resistant gloves must be worn when changing coolant and when handling chemicals or oils

Additional safety precautions

	• When machining abnormal workpieces, a Point of Work (Dynamic) risk assessment MUST be carried out to assess the risks.		• Ensure all guarding is in place and any interlocks are working.
	• Rings and jewelry must not be worn or be covered in work areas.		• Hearing protection should be worn if the indicators are red. Ensure others in the area are complying.
	• Long and loose hair / Loose articles must be contained or secured.		• Where fitted, ensure inbuilt LEV is switched on or use mobile LEV if a lot of smoke or fume is being produced.

Significant hazards



- Entanglement with rotating parts
- Ejection of workpieces or tooling.
- Contact with cutting tools and rotating components.
- Flying swarf and metal chips.
- Manual handling of heavy components.

Lifesaving rules



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Step	Pre-operational safety checks
1	Check workspaces and walkways to ensure no slip/trip hazards are present.
2	Ensure the equipment safety checks have been completed.
3	Check emergency stop and machine controls appear serviceable. Never bypass a safety device
4	Check guards, covers and safety devices are in place and serviceable
5	DC injection brake functional.

Step	Loading, unloading & set-up
1	Open the chuck jaws sufficiently to accept the workpiece.
2	Manoeuvre the workpiece using lifting equipment and appropriate manual handling techniques where required. Where possible, hands-free tooling is to be used to eliminate pinch point and caught between risks.
3	Insert the workpiece fully into the chuck or against any locating stop ensuring there is no back or forth play in the workpiece.
4	Tighten the chuck evenly using the chuck key.
5	Keep the component partially suspended on the crane until satisfied that it has been securely fixed in place using centres and applicable backstops.
6	Position the dial indicator securely on the lathe.
7	Bring the indicator tip into contact with the workpiece.
8	Rotate the chuck SLOWLY by hand only being aware of areas that could be potential pinch points (keyways).
9	Observe the indicator reading and identify any runout.
10	Adjust the workpiece position as required.
11	Retighten the chuck and recheck the reading.
12	Repeat until the workpiece is running within the required tolerance.
13	Perform a final check to ensure all jaws are tight and secure.
14	Remove all tools from the bed and slides of the machine.
15	Remove the chuck key.
16	Remove crane lifting accessories.

Step	Operational safety
1	Ensure correct speed for machining process is selected.
2	Do not try to lift chucks, face plates, or steadies that are too heavy for you – use mechanical aids.
3	Faulty equipment must NOT be used. Immediately report suspect machinery.
4	Machine operator MUST NOT leave the machine unattended while operational and MUST maintain vigilance AT ALL times.
5	Before adjusting or taking measurements, switch off and bring the machine to a complete standstill. Lift the chuck guard and where possible take out of gear to ensure the machine cannot be started. Note: it is not always possible to take lathes out of gear when working with off balance / centre loads.
6	Avoid letting swarf build up on the tool or job. Stop the machine and remove it using a swarf hook or brush. Wear cut-resistant gloves.
7	Always remove the chuck key from the chuck.
8	Do not use the airline to blow away coolant.
9	Ensure all applicable guards are in place while operating the machine.

Step	Housekeeping
1	Switch off the machine and reset all guards to a fully closed position.
2	Leave the machine in a safe, clean, and tidy state.